RYAN SMITH

FULL-STACK SOFTWARE ENGINEER

Chicago, IL · 913-626-8976 · mrryantsmith@gmail.com

mrryantsmith · mrrts · mrryantsmith.vercel.app

Summary

I am a curious and naturally visionary puzzle-solver with a background in theatre and music performance, seeking new opportunities as a full-stack software engineer, especially in the EdTech space. I am fond of working in stacks that include TypeScript and React, love strongly typed, compiled backend languages, and enjoy both functional and object oriented programming. Claude Code and I are a great team.

Experience

Software Engineer
2022 to 2026

Pedago (Quantic School of Business and Technology) • Remote

- Developed full-stack features, for both students and admins, with Rails and Python backend and varied frontend, fixed UX bugs, implemented responsive designs from Sketch and Figma mockups, deployed behind feature flags, reviewed coworkers' pull requests, wrote unit tests and Storybook stories
- Worked with Student Experience team to develop AI experiences and convert legacy AngularJS codebase to React/Redux
- Quickly adapted to new AI tools; used Cursor Agents and Claude Code and completed several large features with quick turnaround; performed independent research to develop original Prompt Engineering techniques; performed self-led data analysis; developed new Claude skills to share with team
- Developed configurations to drive customized UX tailored to a wide range of student/user states
- Implemented DatoCMS-integrated landing pages and many other Gatsby marketing pages on quantic.edu as part of a full rebrand
- Tested our iOS and Android apps in emulators and on-device

Software Developer II
2016 to 2021

Everspring • Chicago, IL

- Developed educational and course authoring software for university online courses, focusing on user experience, scalability, and stakeholder needs, frequently aiming for and achieving 100% unit test coverage and focusing on AA (sometimes AAA) accessibility
- Canvas LMS and LTI: developed several applications, often solo, which securely launched under the Learning Tool Interoperability (LTI) specification and integrated with student and course data via the Canvas API. Sometimes worked with a UX Designer, other times did own design. Steadily matured in usage of back-end architecture design toward a Controller-Service-Repository-Entity pattern.
- Early LTI work had open API access, while later LTIs used a Developer Key and granted specific scoped permissions. Used JWT authentication. Integrated with AWS S3 for file and image storage; back-end image processing and multiple sizes/crops stored in S3. Learned MongoDB for use in all projects.
- First experience with strongly typed languages like Scala (chiefly) and Java, pattern matching, functional programming, dependency injection, and abstraction of repeated CRUD controllers/services/repositories, etc.
- Built custom branded JS/CSS for Canvas clients that allowed Instructional Designers to add rich interactive components directly into course content pages and easily add configuration tables for these components directly into the course page text editor. Focused on accessibility.
- Took initiative to develop several LTIs meant for admin or in-house use (not listed), including one to control permissions to classroom features based on role and institution and also one to provide Instructional Designers with an overview of all classroom components
- Maintained legacy LTI applications with bugfixes and new feature implementations
- Was beginning to self-teach the LTI 1.3 specification for access to deep-linking, line-item grading, OAuth 2.0 + OIDC, names/roles, etc. I have recently worked on an independent Claude Code project for LTI 1.3 As a Service (still in progress)

Volunteer / Worship Leader / Musician / Overseas Missionary / Web Developer
2006 to Present

Various Churches • Chicago, IL

- Participated as part of a rotation of worship leaders / support musicians; selected and learned a set of songs for Sunday services; coordinated bandmate schedules and led rehearsals; played piano; led congregation in prayer and singing at church services; led French worship music for children during four mission trips in France; created and/or maintained several church websites; ran audio/visual station during services

Skills

FULL STACK WEB DEVELOPMENT

HTML/CSS/SASS · TypeScript · Ruby on Rails · SQL · PHP · Git/GitHub · RESTful API Integration · Interm. Photoshop · Agile Development · Scala · Spring · NodeJS/NestJS · React/Redux · Angular · Reactive/RxJs · JQuery · Some Rust · Claude Code · Canvas LMS and LTI · AWS: EC2 and S3 · Custom AI Agents · Unit Tests · NextJS · Responsive Design

Education

Full Stack Web Development Immersive 2016

Dev Bootcamp

BA Theatre 2006

Northwestern University

Projects

AI Advisor / AI Tutor
Python back-end / React front-end. A text and voice chatbot for students to interact with an academic advisor and discuss curriculum and current lesson frame content.

- Implemented chat UI and controls for voice-mode.
- Implemented streaming response tokens over WebSocket connection using reactive programming techniques (Rx).

Student Experience 2.0
Rails back-end / React front-end. Part of a complete overhaul of the in-app student experience, including Admission Application updates, a new Student Dashboard and configurable Sidebar Box system, a complete rebuild of the Student Network, a wide update of app styling, and development of a Slack-like Student Messaging widget. Frequent recent use of Claude Code.

- Worked with team but completed many features solo, often in discussion with C-suite stakeholders.
- Included conversion from legacy AngularJS frontend to React and TypeScript.

WorldThreads — worldthreads.app
Independent project. Rust back-end / React front-end. Built with Claude. AI chat app where users may create fictional worlds and the characters that inhabit them, chat with characters one-on-one or in groups, generate illustrations and videos, generate novel chapters based on conversations, and chat with a Backstage guide. Intended to feel like an "immersive, living storybook".

- Incorporates novel Prompt Engineering techniques emerging from original research: namely, a math-English hybrid notation that defines the chat completion boundaries (based on first principles, fictional world truth, character truth, and present chat moment) using minimal tokens. Feels to me like a promising finding in prompt compression.
- Bring Your Own OpenAI API Key (BYOK); Tauri desktop application

Untitled Player-Piano AI Ecosystem (In Progress)
Independent. Working with the engineers who built a top-tier automated player-piano system installed in grand pianos, including my own. Currently training an AI model to transcribe and play back, live, from any piano audio. Also training on AI analysis of masterworks for eventual AI composition. Built with Claude.

- The ultimate vision is an all-in-one app for recording, composing, sharing, collaborating, and leveraging AI to compose in certain composers' styles and assist in prompt-driven composition.

CourseBuilder
Java Spring back-end / React front-end. A Learning tool (LTI) that operated closely with the Canvas LMS in order to allow academic program heads to quickly bulk create LMS courses with starter content and manage those courses thereafter.

- I was involved early on with the back-end Spring development, particularly in the launch routine for the LTI and the initial route endpoints.
- Afterward, I was the chief developer on the React front-end, which included a custom virtualized, keyboard-navigable data table and many complex forms with reusable form components + tests.

Atlas Publisher
PHP back-end / Angular front-end. Expanded and upgraded the in-house marketing site authoring experience to feel more like a CMS with authoring forms, users, asset management, and version control.

- The experience allowed non-techie users at the universities to manage their own sites without the aid of developers at Everspring.
- The marquee feature was that users could stage new versions of their site and then promote individual pages to the live site once they were approved.

Branching Scenario Creator
Scala/Play back-end / React front-end. LTI that allowed Instructional Designers to visualize and author a choose-your-own-adventure lesson scenario that branched down different narrative paths based on student decisions.

- The main feature was a tree visualization of the branching content flow.
- Made the authoring experience much easier and led to higher usage of the branching scenario in courses.

Skills and Values Portfolio (SVP)
Scala/Play back-end / React front-end. LTI that integrated with course and student data to display student progress through a program of courses, each of which were associated with custom-defined skills and values.

- Students could request feedback for co-curricular activities, see own progress, receive completion badges, and submit periodic reflection essays.
- Program heads and professors could browse student profiles and see summaries of student activity and engagement.

Virtual Community / Jayhawkville
Scala/Play back-end / React front-end. LTI that allowed students to explore a virtual city community by visiting informational pages via a rich map with data overlays. Also included a full fictional school district website students in Sports Management could use as a contextual data-rich laboratory.